

Chapter 13: Deep Learning Frameworks for Text Classification in Low-Resource Languages

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Abstract

Considering the large amount of data or resources all over the internet and most of them are unorganized which increases the complexity to find the exact information and to be able to extract the required information we need to classify data into different categories by the help of a technique called Text Classification. Text classification in low resource like Assamese are which is morphologically rich and agglutinative language. There are very less number of resources present in this field that leads to data scarcity. We will examine the theoretical foundations and some deep learning frameworks starting from traditional models-SVM(Support Vector Machine),RF(Random Forest),CNN,RNN,LSTM to hybrid models-C-LSTM,CNN-GRU,C-LSTM_ATC that can solve the issue for text classifications in low resources languages.

Keywords: Text Classification, Low-Resource Languages, Assamese Language, Deep Learning, AI Models, NLP.

1. Introduction

NLP (Natural Language Processing) is a subset of AI(Artificial Intelligence), CS(Computer Science) and Linguistics.it enable the computer to understand, communicate and process the human language. The Generative AI, models such as LLMs(large language models)-GPT-5, Claude Opus 4.7, Grok 4 now have the skill to understand the human language and create images. NLP is used in chatbots which is used in customer services or smart digital assistants such as Amazon's Alexa, Google's Gemini and Apple's Siri. Computer however process the numerical data. NLP helps to translate the raw human text to structured representation that machine can process.

In today's Digital age, there contain a vast amount of information in the web which is growing in an exponential rate. Millions of documents -such as news, books, blogs, social media post, government record are created every sec because of this unstructured data all over the internet it is very complex to find an specific required information because the data is largely not organized it is extremely difficult to organize file manually which can be time consuming and inefficient.

To able to extract the required information out of it need to be organized in different categories and here comes the role of Text Classification, it is crucial part of the NLP(Natural Language Processing).Text Classification is used for sorting text into predefined label or categories

Unstructured Text Document —► [NLP Text Classifier] —► Assigned Category

Common applications of text classification we use in our day to day life includes:

- i) Spam Filtering: Sorting the e-mail into "spam" or "important".
- ii) Sentiment Analysis: Checking if a customer review is positive, negative, or neutral.
- iii) Topic Sorting: Sorting and categorizing the online news into groups like Sports, Business, or Technology.

iv) Content Moderation: Finding and removing harmful topics and posts on social media

2. Research Challenges in Regional Language

Using the NLP models for regional languages presents major complex challenges. Assamese which is official language of Assam, spoken by over 23 million people it belongs to the Indo Aryan groups, which serves as an example of a low-resource language some key facing challenges are-

a) Morphological Rich Language: Assamese language being an agglutinative language, root words words changes rapidly based on grammer prefixes, suffixes, and case markers, causing a "vocabulary explosion" and vector sparsity.

b) Scarcity of Datasets: Creating these datasets for regional languages takes a lot of time and domain-expert linguists, which is time-consuming process and resource-intensive task.

c) Missing of Standard Tools: Languages like English have lots of software tools for cleaning and organizing text, but regional languages often lack these simple pipelines for this process.

3. Theoretical Foundations and Model Evolution

To address morphological complexity and data sparsity in Assamese language , Text classification frameworks have significantly evolved from statistical traditional classifiers to standalone deep neural networks models to advanced hybrid architectures.

A. Traditional Machine Learning Models

Support Vector Machines (SVM): Traditional machine learning are used to classify rely basic on simple by word counts and statistical feature representations. SVMs construct hyperplanes in high-dimensional space to establish sharp decision boundaries between classes. However, because they rely on bag-of-words or simple frequency counts, they ignore word and gets confused when word shapes changes based on the grammar.

Random Forest (RF): Random Forest is an group of learning approach that builds multiple decision trees during training then combines their outputs to improve the predictive accuracy of the text to categorized. It is similar to SVM it depends on the words counts It cannot read sequential word order which make it bad at understanding the complex grammar structure of the words

B. Foundational Deep Learning Frameworks

i) Recurrent Neural Networks (RNN): RNN read the text in sequential order. It process the text word by word and help them to understand the order of the sequence. But the standard RNNs face a problem in time of training vanishing gradient limitations which is a weak memory make it hard to remember the data during backpropagation, making it difficult as they forget the content across long paragraphs or complicated sentences in the text.

ii) Convolutional Neural Networks (1D-CNN): One-dimensional CNNs use a type of filter which allow them to scan across group of text. They are good at spotting small groups of words or local word patterns, making them highly effective at cleaning up and recognizing of the complex words. However, 1D-CNNs only looks over small group of text and it seems to forget the meaning if the paragraph is long enough.

iii) Long Short-Term Memory (LSTM): LSTMs architectures is designed as a smarter version of the RNN which is introduced to resolve long-range dependency problems faced by NLP. By reading text step-by-step it can remember words using a special memory cell known as gated memory cell structure, it is able to capture effectively and understand full sentence grammar and sequential words. However the limitation arrives when LSTM is applied to small or noisy datasets typical of low-resource languages the gets confused and makes mistakes.

C. Hybrid Deep Learning Architectures

i) C-LSTM Framework: This framework model merges the dual benefits of spatial and sequential feature extraction, researchers combine CNN and LSTM architectures into one unified pipeline. The initial 1D-CNN layer is responsible for the scans of small word groups to filter and clean complex morphological word variations. The extracted feature from first phrase are then passed to an LSTM layer, which reads the sequence step-by-step to understand sentence and overall meaning of the words.

ii) CNN-GRU Framework: It is an alternative hybrid framework that replaces the LSTM module with a memory unit known as GRU(Gated Recurrent Unit). By merging both CNN feature extraction with a GRU's memory unit makes the model quick and effective. The CNN scans the text to find any local pattern and GRU layer reads the feature in a sequence manner. Due to this design it makes it faster and simple than LSTM as it uses few parameter it makes it highly efficient for low-resources system or smaller datasets.

iii) C-LSTM_ATC : It is the most advanced hybrid architecture design model framework by it has integrated Attention (ATC) mechanism layer directly on the top of C-LSTM pipeline. While the CNN handles small group of morphological variations and handle complex words the LSTM layer tracks the sequential sentence grammar step by step, the Attention layer dynamically calculates and assigns importance weights which then is used to highlights the most critical subject words within the text. This design achieves optimal performance since it focus on the relevant information on small datasets without requiring massive amount of computing power.

iv) LSTM-SVM Framework: It is a hybrid model that combines the deep learning model with the traditional machine learning models. First, the LSTM network reads the text step by step to learn the deep meaning of the sentences and how it is link to it. After they finish with a normal neural network output layer, the features learned by the LSTM are passed to the Support Vector Machine (SVM). Then the SVM then makes the final decision by sharply separating the text into different topic categories. This design takes help of LSTM's ability to capture the sequence information and SVM's strength in creating and dividing the topics and clear boundaries between the classes.

3. How the System Works (Methodology)

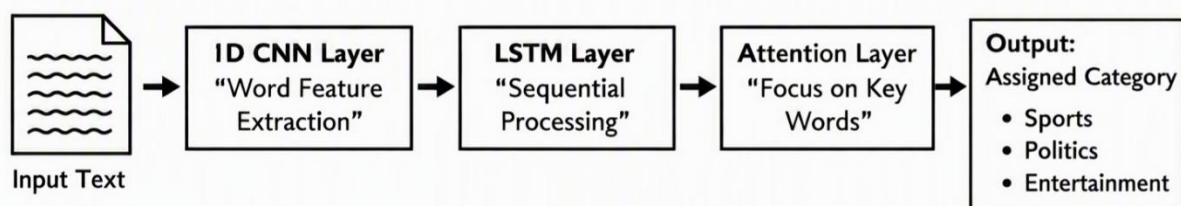


Fig: Workflow of the System

Text Preprocessing

Subword Tokenization: Tools like WordPiece or Byte-Pair Encoding (BPE) split or break the sentences into words and smaller parts. This helps the system recognize the root word and its word meaning even with prefixes or suffixes.

Text Normalization: Unicode normalization ensures consistent encoding that ensure every character is encoded in the same way, while non-standard punctuation like extra symbols and special characters is removed.

Stop-Word Filtering: By removing the filler words such as (and,the,or,is) but keeping important grammar context to preserve sentence structure and context.

4. Feature Extraction

Vector Conversion: The words are converted to numeric data that represent its meaning known as dense vectors.

FastText Embeddings: It uses the character *n-grams* which is small letter group to represent words that tends to change form with adding prefix or suffix to the text.

Text Recognition: This allows the system to understand and learn the new word forms with its meaning for better words recognition.

5. Classification Pipeline

The word vectors are passed through hybrid neural layers:

CNN: It can detect local word patterns such as prefix, suffix or short words.

Recurrent layer (LSTM/GRU): It is used captures the sentence grammar and the word order and sequences.

Attention mechanism: It is used to assign weight and to highlights the most important words.

Softmax/SVM: It is used to makes the final category list by dividing the text or words into different topics (Sports, Polity, Entertainment).

Framework	Strenghts	Weakness
CNN-GRU	Model is efficient, lesser parameters, less training time	As GRU uses only two gates, it can miss subtle context in morphologically complex language such as Assamese
C-LSTM-ACT	Combines advantages of CNN that can find local features and LSTM with its strong long-term memory. Also, Attention	Mechanism that is used is Computationally expensive but better at handling complex grammars

Why do CNN-GRU model framework is not as effective compared to C-LSTM-ATC for Assamese text classification model framework?

Morphology Words in Assamese often depend on suffixes and prefixes. CNN can detect local changes but GRU with simpler gates may not preserve long-range dependencies. Long sentences Many Assamese texts contain sentences with complex structure. It means that GRU will forget the beginning of the sentence and will not detect grammar. Low resources data The data about Assamese is not too big. GRU might not work properly with this kind of data while C-LSTM-ATC would be able to do it. Attention mechanism in C-LSTM-ATC. In ATC (Attention) layer the model can pay attention to most important words. CNN-GRU does not have this mechanism and it cannot highlight important words in classification.

CNN-GRU is highly efficient and light but loses out in terms of depth of context. C-LSTM-ATC is heavy but way better for Assamese since it can deal with more complicated word forms and is able to preserve long-range sentence meaning.

6. Expected Result

- i) Improved Word Difficulties Handling: By splitting up difficult words into smaller pieces, the model is able to efficiently parse out Assamese rare or modified words.
- ii) Smarter Identification of Important Keywords: The Attention Highlighter makes it easy for the system to identify important subject keywords.
- iii) Fast and Effective with Less Computation: The C-LSTM-ATC algorithm performs efficiently even in small data sets without expensive computing devices.

Future Scope

- i) Creating Quality Datasets: There is a requirement of developing quality datasets and some basic software tools for Assamese and neighboring regional languages.

ii) Developing Light-Weight AI Models: There could be some efforts made in trying to adapt modern light-weight transformer models that work well in low-resource regional languages.

iii) Practical Application: The intelligent AI models developed can be practically used in applications like local news applications, comments filtering on social media sites, etc.

7. Conclusion

It is hard to sort out text in languages such as Assamese with difficult grammatical forms and scarcity of digitized data. The previous algorithms face difficulty, but using multiple AI techniques in hybrid deep learning models can solve this problem. With CNNs for filtering variations in words, LSTMs to capture semantic information in sentences, and Attention to focus on relevant words, the C-LSTM-ATC model is a perfect approach.

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